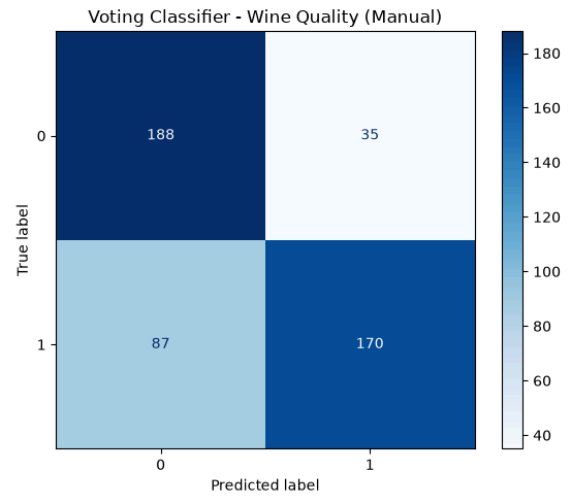
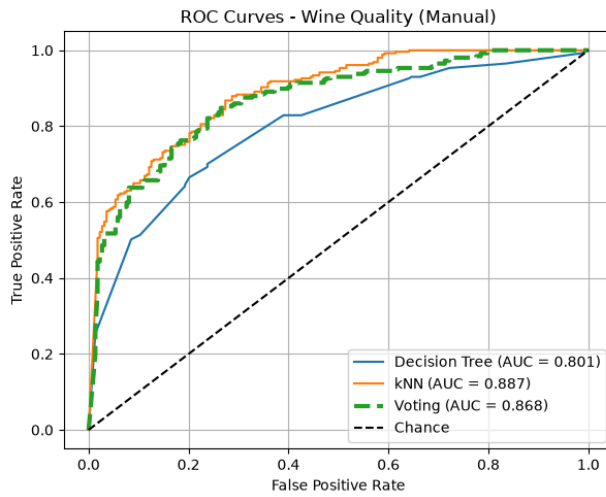


Merwyn Prince Lobo
PES2UG24CS276
E

The screenshot displays a Jupyter Notebook titled 'Week4_Lab_Boilerplate.ipynb' within the Visual Studio Code editor. The notebook is running on a Python 3.14.4 environment. The code in the notebook performs the following steps:

- PROCESSING DATASET: WINE QUALITY**: Loads and preprocesses the wine quality dataset. The training set shape is (1119, 11) and the testing set shape is (480, 11).
- RUNNING MANUAL GRID SEARCH FOR WINE QUALITY**:
 - Manual Grid Search for Decision Tree**: Finds the best parameters for a Decision Tree classifier, resulting in a best cross-validation AUC of 0.7876.
 - Manual Grid Search for kNN**: Finds the best parameters for a k-Nearest Neighbors classifier, resulting in a best cross-validation AUC of 0.8766.
- EVALUATING MANUAL MODELS FOR WINE QUALITY**:
 - Manual Voting Classifier**: Evaluates a voting classifier, showing a performance of Accuracy: 0.7458, Precision: 0.8293, Recall: 0.6615, F1: 0.7359, and AUC: 0.8677.

The output of the notebook is truncated, with a message indicating that the full output can be viewed in a scrollable element or a text editor.



```

File Edit Selection View Go Run Terminal Help Machine_Learning_Lab
Week4_Lab_Boilerplate.ipynb M X
lab-4 > Week4_Lab_Boilerplate.ipynb > M4 Models and Parameter Grids > M4 Part 9: Execute the Complete Lab > # --- Run Pipeline for All Datasets ---
Generate + Code + Markdown Run All Restart Clear All Outputs Jupyter Variables Outline
...
=====
RUNNING BUILT-IN GRID SEARCH FOR WINE QUALITY
=====
--- GridSearchCV for Decision Tree ---
Best parameters: {'classifier__max_depth': 5, 'classifier__min_samples_split': 10, 'scaler': MinMaxScaler(), 'selector_k': 4}
Best cross-validation AUC: 0.7876
Execution time: 6.54 seconds

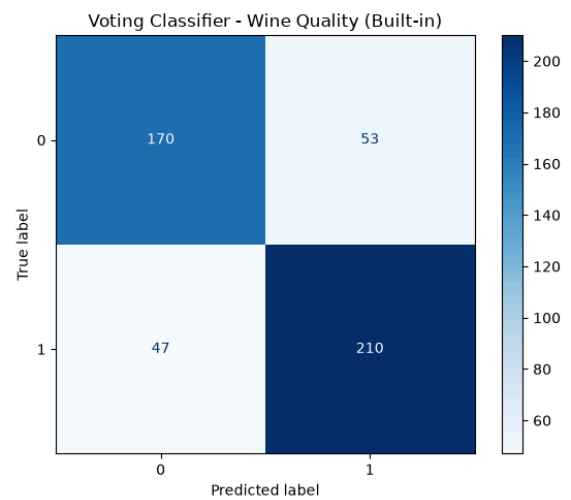
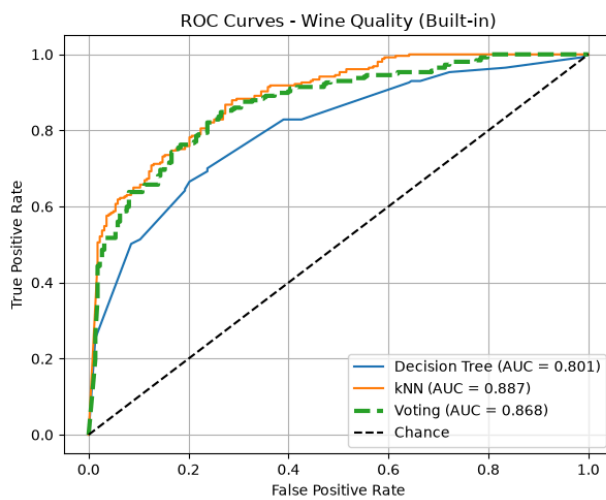
--- GridSearchCV for kNN ---
Best parameters: {'classifier__n_neighbors': 9, 'classifier__p': 1, 'classifier__weights': 'distance', 'scaler': MinMaxScaler(), 'selector_k': 4}
Best cross-validation AUC: 0.8766
Execution time: 2.63 seconds

=====
EVALUATING BUILT-IN MODELS FOR WINE QUALITY
=====
--- Individual Model Performance ---

Decision Tree:
Accuracy: 0.7292
Precision: 0.7725
Recall: 0.7004

...
--- Built-in Voting Classifier ---
Voting Classifier Performance:
Accuracy: 0.7917, Precision: 0.7985
Recall: 0.8171, F1: 0.8077, AUC: 0.8677
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...
...
ROC Curves - Wine Quality (Built-in) Voting Classifier - Wine Quality (Built-in)
main* 0 0 0 21:32 28-08-2026

```



```
File Edit Selection View Go Run Terminal Help Machine_learning_lab

Week4_Lab_Boilerplate.ipynb M X
lab-4 > Week4_Lab_Boilerplate.ipynb > M4 Models and Parameter Grids > M4 Part 9: Execute the Complete Lab > # --- Run Pipeline for All Datasets ---
Generate Code Markdown Run All Restart Clear All Outputs Jupyter Variables Outline .venv (3.14.4.final.0) (Python 3.14.4)

=====
RUNNING RANDOMIZED SEARCH FOR WINE QUALITY
=====

--- RandomizedSearchCV for Decision Tree ---
Best parameters: {'classifier_max_depth': 12, 'classifier_min_samples_split': 9, 'scaler': StandardScaler(), 'selector_k': 4}
Best cross-validation AUC: 0.7872
Execution time: 0.19 seconds

--- RandomizedSearchCV for kNN ---
Best parameters: {'classifier_n_neighbors': 16, 'classifier_p': 1, 'classifier_weights': 'distance', 'scaler': MinMaxScaler(), 'selector_k': 3}
Best cross-validation AUC: 0.8731
Execution time: 0.24 seconds

=====
EVALUATING RANDOMIZED MODELS FOR WINE QUALITY
=====

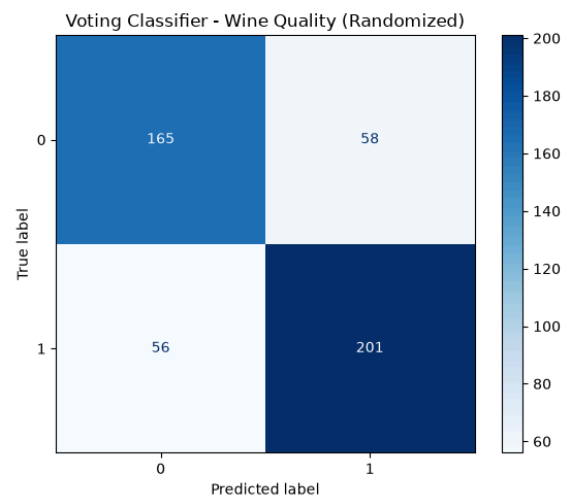
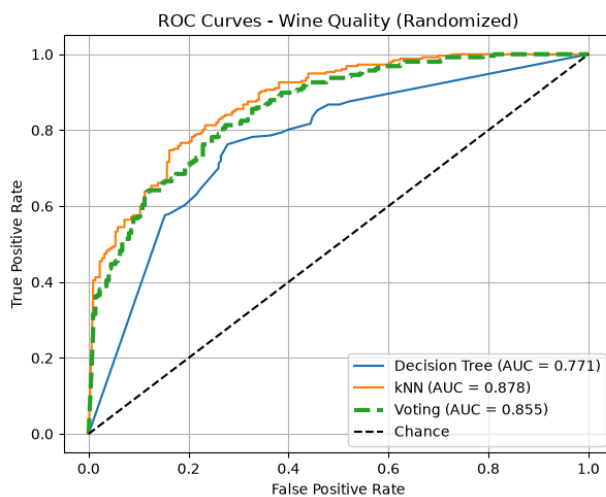
--- Individual Model Performance ---

Decision Tree:
Accuracy: 0.7333
Precision: 0.7611
Recall: 0.7315

---
--- Randomized Voting Classifier ---
Voting Classifier Performance:
Accuracy: 0.7625, Precision: 0.7761
Recall: 0.7821, F1: 0.7791, AUC: 0.8554

Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...

Spaces: 4 {} Cell 28 of 28
21:32
28-08-2026
```



```
File Edit Selection View Go Run Terminal Help Machine_learning_lab

Week4_Lab_Boilerplate.ipynb M X
lab-4 > Week4_Lab_Boilerplate.ipynb > M4 Models and Parameter Grids > M4 Part 9: Execute the Complete Lab > # --- Run Pipeline for All Datasets ---
Generate + Code + Markdown | Run All | Restart | Clear All Outputs | Jupyter Variables | Outline ...

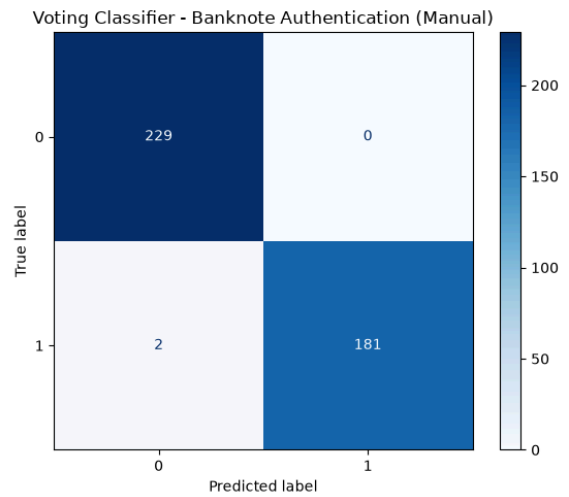
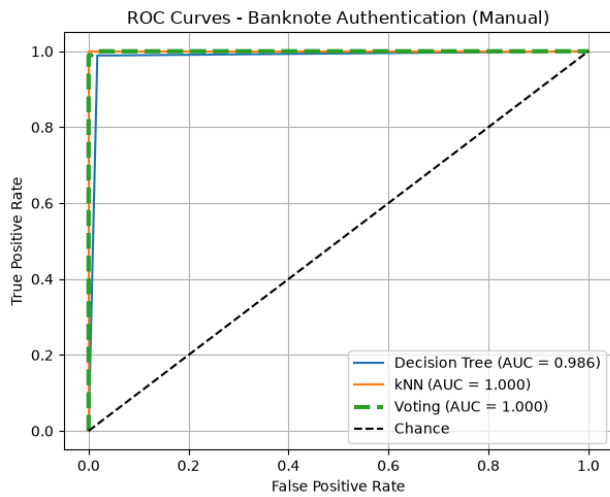
Jupyter Notebook
.venv (3.14.4.final.0) (Python 3.14.4)

...
completed processing for Wine Quality
=====
PROCESSING DATASET: HR ATTRITION
=====
HR Attrition dataset not found. Please place 'WA_Fn-UseC_HR-Employee-Attrition.csv' inside a 'data/' folder.
Skipping HR Attrition due to loading error.

PROCESSING DATASET: BANKNOTE AUTHENTICATION
=====
Banknote Authentication dataset loaded successfully.
Training set shape: (960, 4)
Testing set shape: (412, 4)
=====

RUNNING MANUAL GRID SEARCH FOR BANKNOTE AUTHENTICATION
=====
--- Manual Grid Search for Decision Tree ---
Best parameters for Decision Tree: {'scaler': StandardScaler(), 'selector_k': 3, 'classifier_max_depth': 10, 'classifier_min_samples_split': 10}
Best cross-validation AUC: 0.9869
...
--- Manual Voting Classifier ---
Voting Classifier Performance:
Accuracy: 0.9951, Precision: 1.0000
Recall: 0.9891, F1: 0.9945, AUC: 0.9999
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...

ROC Curves - Banknote Authentication (Manual) Voting Classifier - Banknote Authentication (Manual)
```



```
File Edit Selection View Go Run Terminal Help Machine_learning_lab

Week4_Lab_Boilerplate.ipynb M X
lab-4 > Week4_Lab_Boilerplate.ipynb > M4 Models and Parameter Grids > M4 Part 9: Execute the Complete Lab > # --- Run Pipeline for All Datasets ---
Generate + Code + Markdown | Run All | Restart | Clear All Outputs | Jupyter Variables | Outline ...

Jupyter Notebook
Python 3.14.4.final.0 (Python 3.14.4)

...
=====
RUNNING BUILT-IN GRID SEARCH FOR BANKNOTE AUTHENTICATION
=====

--- GridSearchCV for Decision Tree ---
Best parameters: {'classifier__max_depth': 10, 'classifier__min_samples_split': 10, 'scaler': StandardScaler(), 'selector__k': 3}
Best cross-validation AUC: 0.9869
Execution time: 1.19 seconds

--- GridSearchCV for KNN ---
Best parameters: {'classifier__n_neighbors': 5, 'classifier__p': 2, 'classifier__weights': 'distance', 'scaler': StandardScaler(), 'selector__k': 3}
Best cross-validation AUC: 1.0000
Execution time: 1.58 seconds

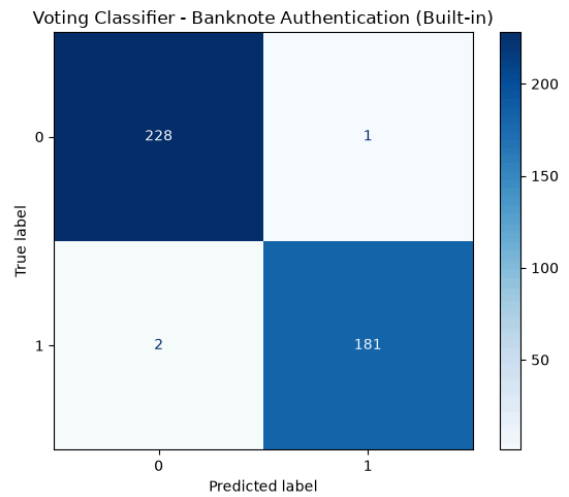
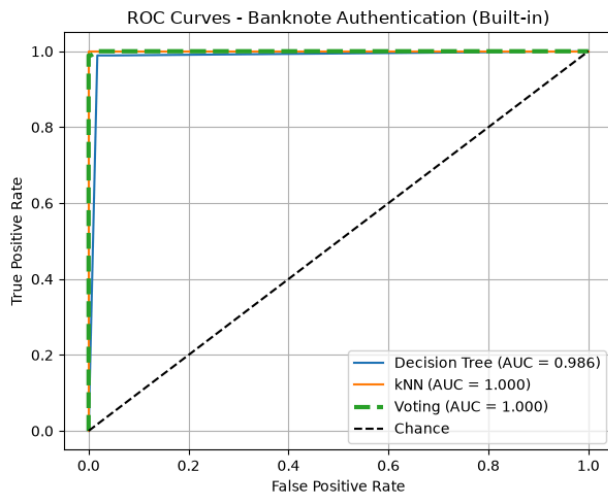
=====
EVALUATING BUILT-IN MODELS FOR BANKNOTE AUTHENTICATION
=====

--- Individual Model Performance ---

Decision Tree:
Accuracy: 0.9854
Precision: 0.9784
Recall: 0.9891
...

--- Built-in Voting Classifier ---
Voting Classifier Performances:
Accuracy: 0.9927, Precision: 0.9945
Recall: 0.9891, F1: 0.9918, AUC: 0.9999
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...

=====
ROC Curves - Banknote Authentication (Built-in) Voting Classifier - Banknote Authentication (Built-in)
=====
```



```
File Edit Selection View Go Run Terminal Help Machine_learning_lab

Week4_Lab_Boilerplate.ipynb M X
lab-4 > Week4_Lab_Boilerplate.ipynb > M4 Models and Parameter Grids > M4 Part 9: Execute the Complete Lab > # --- Run Pipeline for All Datasets ---
Generate Code Markdown Run All Restart Clear All Outputs Jupyter Variables Outline .venv (3.14.4.final.0) (Python 3.14.4)

...

=====
RUNNING RANDOMIZED SEARCH FOR BANKNOTE AUTHENTICATION
=====

--- RandomizedSearchCV for Decision Tree ---
Best parameters: {'classifier__max_depth': 8, 'classifier__min_samples_split': 11, 'scaler': StandardScaler(), 'selector_k': 3}
Best cross-validation AUC: 0.9869
Execution time: 0.15 seconds

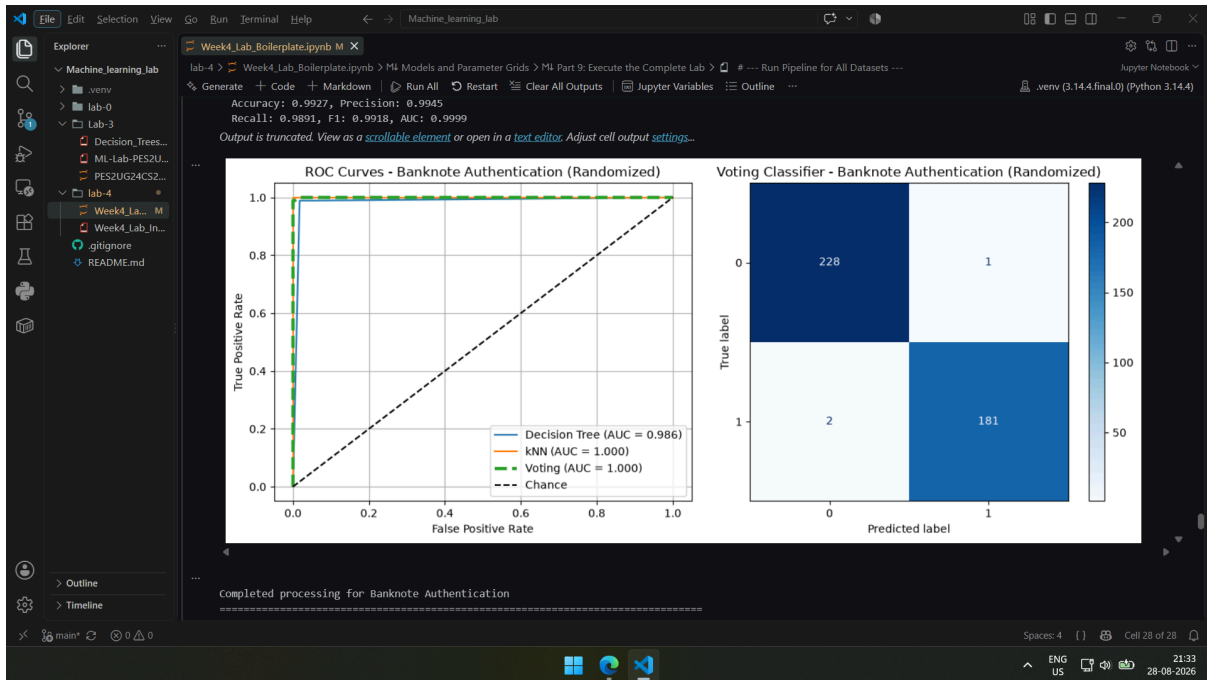
--- RandomizedSearchCV for kNN ---
Best parameters: {'classifier__n_neighbors': 16, 'classifier__p': 1, 'classifier__weights': 'distance', 'scaler': MinMaxScaler(), 'selector_k': 3}
Best cross-validation AUC: 1.0000
Execution time: 0.13 seconds

=====
EVALUATING RANDOMIZED MODELS FOR BANKNOTE AUTHENTICATION
=====

--- Individual Model Performance ---

Decision Tree:
Accuracy: 0.9854
Precision: 0.9784
Recall: 0.9891
...
--- Randomized Voting Classifier ---
Voting Classifier Performance:
Accuracy: 0.9927, Precision: 0.9945
Recall: 0.9891, F1: 0.9918, AUC: 0.9999
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...

Spaces: 4 {} Cell 28 of 28
21:33
28-08-2026
```



```
File Edit Selection View Go Run Terminal Help Machine_learning_lab

Week4_Lab_Boilerplate.ipynb M X
lab-4 > Week4_Lab_Boilerplate.ipynb > M4 Models and Parameter Grids > M4 Part 9: Execute the Complete Lab > # --- Run Pipeline for All Datasets ---
Generate Code Markdown Run All Restart Clear All Outputs Jupyter Variables Outline .venv (3.14.4.final.0) (Python 3.14.4)

False Positive Rate Predicted label

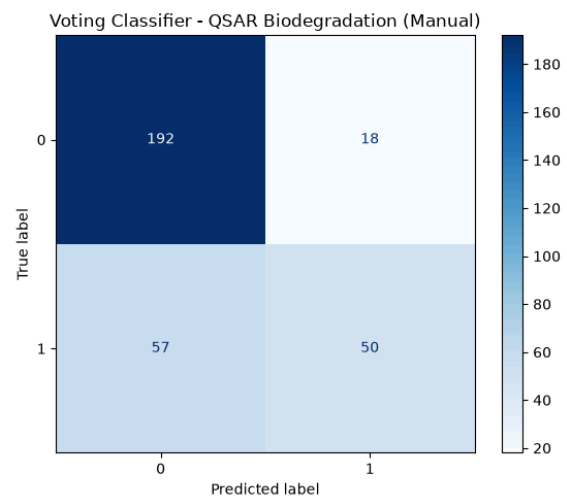
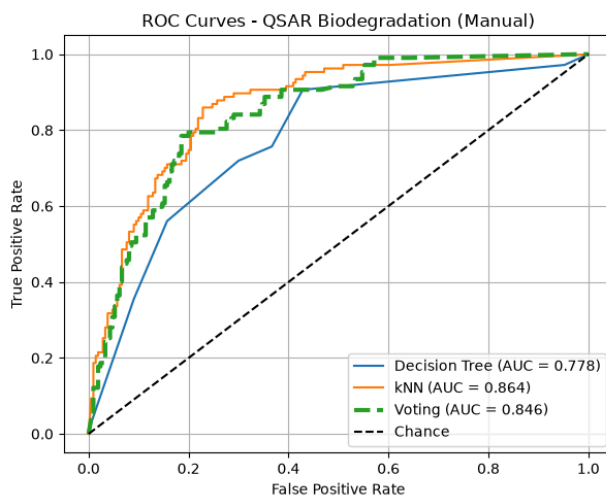
...

Completed processing for Banknote Authentication
=====
PROCESSING DATASET: QSAR BIODEGRADATION
=====
QSAR Biodegradation dataset loaded successfully.
Training set shape: (738, 41)
Testing set shape: (317, 41)
=====

RUNNING MANUAL GRID SEARCH FOR QSAR BIODEGRADATION
=====
--- Manual Grid Search for Decision Tree ---
-----
Best parameters for Decision Tree: {'scaler': StandardScaler(), 'selector_k': 10, 'classifier_max_depth': 3, 'classifier_min_samples_split': 2}
Best cross-validation AUC: 0.8192
--- Manual Grid Search for kNN ---
-----
Best parameters for kNN: {'scaler': MinMaxScaler(), 'selector_k': 10, 'classifier_n_neighbors': 9, 'classifier_weights': 'distance', 'classifier_p': 1}
Best cross-validation AUC: 0.8807
=====

--- Manual Voting Classifier ---
Voting Classifier Performance:
Accuracy: 0.7634, Precision: 0.7353
Recall: 0.4673, F1: 0.5714, AUC: 0.8464
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...

Spaces: 4 {} Cell 28 of 28
21:33
28-08-2026
```



```
File Edit Selection View Go Run Terminal Help Machine_learning_lab

Week4_Lab_Boilerplate.ipynb M X
lab-4 > Week4_Lab_Boilerplate.ipynb > M4 Models and Parameter Grids > M4 Part 9: Execute the Complete Lab > # --- Run Pipeline for All Datasets ---
Generate Code Markdown Run All Restart Clear All Outputs Jupyter Variables Outline .venv (3.14.4.final.0) (Python 3.14.4)

=====
RUNNING BUILT-IN GRID SEARCH FOR QSAR BIODEGRADATION
=====

--- GridSearchCV for Decision Tree ---
Best parameters: {'classifier__max_depth': 3, 'classifier__min_samples_split': 2, 'scaler': StandardScaler(), 'selector__k': 10}
Best cross-validation AUC: 0.8192
Execution time: 2.69 seconds

--- GridSearchCV for KNN ---
Best parameters: {'classifier__n_neighbors': 9, 'classifier__p': 1, 'classifier__weights': 'distance', 'scaler': MinMaxScaler(), 'selector__k': 10}
Best cross-validation AUC: 0.8807
Execution time: 3.83 seconds

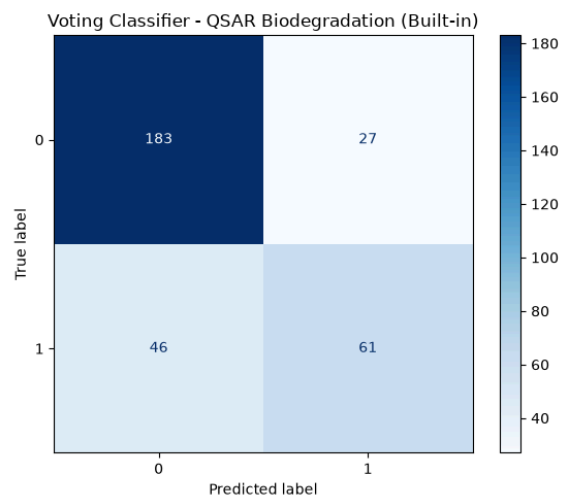
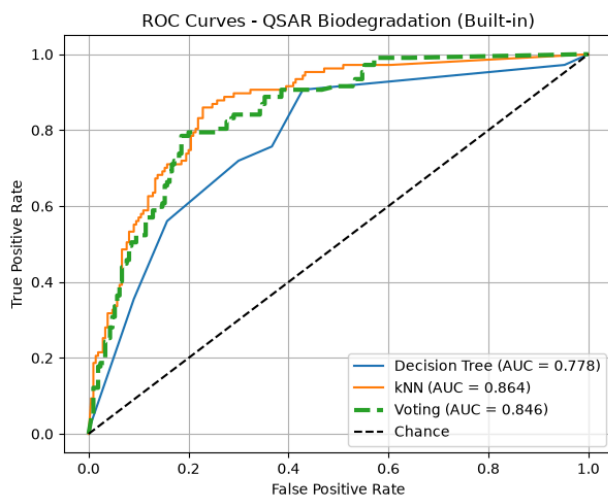
=====
EVALUATING BUILT-IN MODELS FOR QSAR BIODEGRADATION
=====

--- Individual Model Performance ---

Decision Tree:
Accuracy: 0.7476
Precision: 0.6452
Recall: 0.5607
...

--- Built-in Voting Classifier ---
Voting Classifier Performance:
Accuracy: 0.7697, Precision: 0.6932
Recall: 0.5701, F1: 0.6256, AUC: 0.8464
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...

Spaces: 4 () Cell 28 of 28
ENG US 21:33 28-08-2026
```



The screenshot displays a Jupyter Notebook titled "Week4_Lab_Boilerplate.ipynb" within a "Machine_Learning_Lab" environment. The notebook is running on a ".venv (3.14.4.final.0) (Python 3.14.4)" kernel. The left sidebar shows the file explorer with folders "lab-0", "lab-3", "lab-4", and "Week4_Lab_Boilerplate.ipynb". The main area shows the notebook content, which includes a section titled "RUNNING RANDOMIZED SEARCH FOR QSAR BIODEGRADATION". This section contains two sub-sections: "RandomizedSearchCV for Decision Tree" and "RandomizedSearchCV for kNN". The Decision Tree section shows best parameters: {'classifier__max_depth': 12, 'classifier__min_samples_split': 11, 'scaler': MinMaxScaler(), 'selector__k': 28}, with a best cross-validation AUC of 0.8684 and an execution time of 0.30 seconds. The kNN section shows best parameters: {'classifier__n_neighbors': 17, 'classifier__p': 1, 'classifier__weights': 'uniform', 'scaler': StandardScaler(), 'selector__k': 28}, with a best cross-validation AUC of 0.9085 and an execution time of 0.26 seconds. Below these, there is a section titled "EVALUATING RANDOMIZED MODELS FOR QSAR BIODEGRADATION" which includes "Individual Model Performance" for a Decision Tree (Accuracy: 0.7950, Precision: 0.6909, Recall: 0.7103) and a "Randomized Voting Classifier" (Accuracy: 0.8265, Precision: 0.7364, Recall: 0.7570, F1: 0.7465, AUC: 0.9052). The notebook also includes a footer with "Spaces: 4", "Cell 28 of 28", and a timestamp of "21:33 28-08-2026".

```
lab-4 > Week4_Lab_Boilerplate.ipynb > M4 Models and Parameter Grids > M4 Part 9: Execute the Complete Lab > # --- Run Pipeline for All Datasets ---
Generate Code Markdown Run All Restart Clear All Outputs Jupyter Variables Outline
---
=====
RUNNING RANDOMIZED SEARCH FOR QSAR BIODEGRADATION
=====
--- RandomizedSearchCV for Decision Tree ---
Best parameters: {'classifier__max_depth': 12, 'classifier__min_samples_split': 11, 'scaler': MinMaxScaler(), 'selector__k': 28}
Best cross-validation AUC: 0.8684
Execution time: 0.30 seconds

--- RandomizedSearchCV for kNN ---
Best parameters: {'classifier__n_neighbors': 17, 'classifier__p': 1, 'classifier__weights': 'uniform', 'scaler': StandardScaler(), 'selector__k': 28}
Best cross-validation AUC: 0.9085
Execution time: 0.26 seconds

=====
EVALUATING RANDOMIZED MODELS FOR QSAR BIODEGRADATION
=====

--- Individual Model Performance ---

Decision Tree:
Accuracy: 0.7950
Precision: 0.6909
Recall: 0.7103

---
--- Randomized Voting Classifier ---
Voting Classifier Performance:
Accuracy: 0.8265, Precision: 0.7364
Recall: 0.7570, F1: 0.7465, AUC: 0.9052

Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings.
---
=====
```

